

The influence of demand and supply side pressures on the quantity and quality of inventive activity

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I. INTRODUCTION

It is now widely accepted that economic factors have a role to play in the determination of the rate and direction of technological change (see Nelson, 1962). One unresolved aspect of the research concerns the relative importance of demand and supply pressures on the amount of inventive activity. Schmookler (1966) argued that demand-pull factors affect the anticipated revenues from inventive activity, thereby influencing the number and distribution of inventions coming onto the market. Little further empirical work appeared until a recent article by Stoneman (1979) which reports on a model incorporating both demand and cost variables, distinguishing the role played by each in determining the level of inventive activity. Tests using data for UK manufacturing industries appeared to confirm the importance of demand-pull influences, but also, indicated a key role for supply-side factors. There are, however, a number of theoretical and empirical problems with this earlier study and this paper therefore extends the theoretical model and tests the revised specification on a more compatible data set. The new results largely reaffirm the earlier conclusions about the importance of supply-side considerations, but they clearly indicate the need to examine the supply of inventions in a more dynamic context and also allow for variations in the quality of inventive output.

II. A MODEL OF INVENTIVE ACTIVITY

Stoneman model

It is useful to begin by outlining the Stoneman (1979) model, which is based on profit maximizing behaviour by inventing firms. Ignoring lags for the moment, this requires the firm to maximize,

$$\Pi_{ij} = x_{ij}R_{ij} - C_{ij}(x_{ij}) \quad (1)$$

where Π denotes profits from the invention, x is the number of inventions proxied by the number of patents,¹ R is the average revenue per invention, $C(x)$ is the cost of producing inventions which is a function of x , i denotes the i th firm and j the j th industry. It is not possible to measure R directly, but it is postulated that,

$$R_{ij} = R_j = \alpha_j \left(\frac{I}{Y} \right)_j^{\xi_j} Y_j^{\gamma_j} x_j^{\eta_j} \quad (2)$$

where α , ξ and γ are constants, I denotes investment, Y is the level of output, and η_j is the elasticity of expected revenue from a patent in industry j with respect to the number of patents produced in industry j , $-1 < \eta_j < 0$. The average revenue per invention is assumed to be the same for both the firm and the industry. Movements in I/Y and Y represent variations in the availability of profitable outlets for inventions. It is assumed that the firm maximizes profits based on Equation 1. Then, on substitution of Equation 2 into the resulting first-order condition and on the further assumptions that conjectural variations by firm i are constant and the marginal cost per invention is equal to the average cost, $C_j = (RD/x)_j$, it can be shown that,

$$\log x_j = \Lambda_j + \frac{1}{\eta_j} \log \left(\frac{RD}{x} \right)_j - \frac{\xi_j}{\eta_j} \log \left(\frac{I}{Y} \right)_j - \frac{\gamma_j}{\eta_j} \log Y_j \quad (3)$$

where Λ_j is a constant. Thus the equation to be estimated is comprised of both cost and demand influences.

Extensions to the model

Data and lags in the system. Stoneman considered the lagged effects of I/Y and Y on x in some detail, with the empirical results supporting a simple, one period lag. However, little or no consideration was given to the appropriate lag on the RD/x variable. The lack of information about R & D made the testing of such a lag impossible in the earlier study. In the data used below, this lag assumes a much greater importance and attention is therefore paid to its likely magnitude.

The data used in the Stoneman study are taken from Boehm (1967). They refer to patent acceptances which, given the procedural mechanisms of the Patent Office,² involved an average time between application and acceptance of about two years. Thus, R & D should be lagged by at least two years relative to the patent variable in constructing the average cost per invention, RD/x . In addition, however, there is still the lag between the expenditure on R & D and the application for a patent to be considered.³ Given information about the procedural lag and an estimate of the overall lag, it should be possible to calculate the lag between expenditure and

¹For reviews of the various problems associated with the use of patent data, see Sanders (1962), Kuznets (1962), Bosworth (1973), Freeman (1974) and Bosworth (1984). It is certainly not possible to 'play-down' the special nature of the data (Stoneman, 1979, p. 385 and p. 391).

²The procedural mechanisms for obtaining a patent have been substantially altered by the Patent Act (1977) but these do not affect the sample periods adopted in either study.

³Bosworth (1973, p. 98)

application. Estimates of the overall lag, however, are affected by the variability of the lags and the general problems associated with modelling lags.

The stream of net revenues from invention. The use of Equation 2 as an approximation to the net revenue stream associated with the invention appears suspect insofar as I/Y and Y relate to a single period, $t - 1$. While it may be argued that firms faced by an uncertain future adopt rules of thumb, it seems unlikely that they will take no account of the nature of this stream over time. In this paper, therefore, a compromise is adopted, using a rule of thumb in a discounted cash flow procedure. The revenue stream is now written as

$$R'_t = \sum_{t=0}^P \frac{\alpha_t \left(\frac{I}{Y}\right)_t^\xi Y_t^\gamma x_t^\eta}{(1+i)^t} \quad (4)$$

where the firm and industry subscripts are omitted for simplicity of exposition and where t denotes the year, P is the expected life of the patented invention, and i is the rate of discount. The variables with subscript t are meant to denote their value as perceived by the firm at time t . In practice, this might be the actual value of the variable in the previous period. Little or no progress can be made unless assumptions are made about expectations with regard to movements in I , Y and x . At this stage, the assumption is made that their perceived values at time t are, as a rule of thumb, thought likely to continue for P periods ahead. Using R to denote Stoneman's estimate of revenues, it is possible to write,

$$R' = ZR = \left[\frac{(1+i)^P - 1}{i(1+i)^P} \right] R \quad (5)$$

where Z can be constructed given information about P and i ; $\partial R'/\partial Z \geq 0$; and R is defined by Equation 2.

One problem with Equation 4 is that α , ξ , γ and η may not be independent of t or P . Attempting to allow for this directly leads to functional forms that are unlikely to be estimated from existing data. There is also the problem that firms may be more uncertain about future than current flows. They may therefore give a smaller weighting to such flows in their investment decisions. As a first approximation the coefficient of Z can be allowed to vary from unity, $R' = Z^\theta R$, with the expectation that $0 < \theta \leq 1$.

Considerable discussion has taken place in the literature indicating that imperfect information and the uncertainty surrounding the invention and innovation processes make the supply of internally generated funds within the firm a potentially important determinant of the level of these sorts of discretionary investments.⁴ Thus, a perfectly elastic supply of funds does not exist for inventive and innovative activities at the going market rate of interest. The appropriate discount rate i becomes

$$i = i(r, \Pi_E) \quad (6)$$

where r is the market rate of interest and Π_E denotes the economic profit of the firm, reflecting the supply of internally generated funds available for discretionary investment purposes.

⁴See for example, Grabowski (1968), Mueller (1967) and Smyth *et al.* (1972).

While this argument appears fairly plausible, it seems likely that the role played by profits may be more complicated than this would suggest. There are strong grounds, for example, for believing that inventive activity may be more difficult to finance from external sources than innovative activity. This suggests that the amount of R & D that can be undertaken by the firm will be correlated with profits and will therefore be of approximately the same period, affecting the number of inventions appearing on the market around three periods late. On the other hand, we might expect the innovation to be less influenced by the availability of profits. Thus, the availability of 'internally generated funds' when the firm is taking out the patent may be a less important influence. Indeed, there are grounds for believing that high profits at the time of the innovation decision may discourage further innovative activity: the firm may already hold a portfolio of patents that act as an effective barrier to entry and high profit performance on existing technology may reduce the net returns to the introduction of new technology (Arrow, 1962).

Leaving aside for the moment the question of the appropriate lag, a solution based on Equation 5, would result in a complicated functional form for the regression equation. Hence, an *ad hoc* specification for the revenue function is adopted,

$$R' = \alpha \left(\frac{I}{Y} \right)^\xi Y^\gamma x^\eta P^{\theta_1} r^{\theta_2} \Pi_E^{\theta_3} \quad (7)$$

$$\text{where } \frac{\partial R'}{\partial P} > 0, \quad \frac{\partial R'}{\partial r} < 0 \quad \text{and} \quad \frac{\partial R'}{\partial \Pi_E} \geq 0.$$

Quality of inventive output. The problems of using a simple patent count have been recognized for many years where inventions vary in their technical and commercial quality.⁵ This paper adopts a suggestion that commercial success will be reflected in the length of patent life P .⁶ In addition, however, it seems likely that the commercial importance of any given patented idea will depend on how widely the invention is used. In general, commercially more important inventions can be expected to be patented in a larger number of countries A .

The DCF approach already takes variations in the commercial life of the patent into account. Inclusion of the number of countries of use involves a further slight modification,

$$R'' = \alpha \left(\frac{I}{Y} \right)^\xi Y^\gamma x^\eta Z^\theta A^\rho \quad (8)$$

where $\frac{\partial R''}{\partial P} > 0$ and $\frac{\partial R''}{\partial A} > 0$. The production of higher quality inventions is not without cost,⁷ however, and the total cost function is therefore, rewritten,

$$C'' = RD = \beta x^\zeta P^\phi A^\tau \quad (9)$$

where β, ζ, ϕ and τ are constants. Total profits from invention are the sum of total revenues xR''

⁵ Sanders *et al.* (1958) and Boyer (1970).

⁶ Bosworth (1973), see also Federico (1958) and Nordhaus (1969).

⁷ The work of Griliches (1960), for example, emphasized the associated costs of adapting hybrid corn to different climatic regions and effectively, therefore, to different markets.

minus total costs of the invention. Hence, the first-order condition for a maximum with regard to x yields

$$\log x = \frac{1}{\eta} \frac{\zeta}{(\eta + 1)} \log \left(\frac{RD}{x} \right) - \frac{1}{\eta} \log \alpha - \frac{\zeta}{\eta} \log \frac{I}{Y} - \frac{\gamma}{\eta} \log Y - \frac{\theta}{\eta} \log Z - \frac{\rho}{\eta} \log A \quad (10)$$

where $-1 < \eta < 0$ and hence $[1/\eta(\eta + 1)] < 0$. Thus the ratio of the remaining coefficients *a priori* possess positive signs overall. Similar equations can be isolated for P and A .

Clearly estimation of Equation 10 alone will lead to statistical problems because P and A are not given exogeneously. In principle, this problem is avoided by solving the simultaneous equations for x , P and A in terms of the independent variables RD , I/Y , Y , etc. In practice, because the discounting procedure leads to jointly multiplicative and additive terms in i and P , the mathematics become extremely complex. An *ad hoc* specification, is therefore adopted,

$$R'' = \alpha \left(\frac{I}{Y} \right)^{\zeta} Y^{\gamma} x^{\eta} P^{\theta_1} i^{\theta_2} A^{\rho} \quad (11)$$

where $\theta_1 > 0$ and $\theta_2 < 0$. Thus, Equation 10 and the analogous equations for P and A (which are not shown) can be rewritten, replacing $(\theta/\eta) \log Z$ by $(\theta_1/\eta) \log P + (\theta_2/\eta) \log i$. The simultaneous system can now be solved, and the solution for x appears as

$$x = \kappa \left(\frac{I}{Y} \right)^{-\zeta/\beta} Y^{-\gamma/\beta} i^{-\theta_2/\beta} \left(\frac{RD}{P} \right)^{\theta_1/\beta} \left(\frac{RD}{A} \right)^{\rho/\beta} \left(\frac{RD}{x} \right)^{(1-\theta_1-\rho)/\beta} \quad (12)$$

where $\beta = (\eta + \rho + \theta_1)$ and κ is a constant. RD/P , RD/A and RD/x are the average costs of the various dimensions of inventive activity which are given exogeneously.

Domestic and foreign inventions. A fundamental problem with the acceptance data used in the Stoneman study is what it relates to all patents, whether of British or foreign origin.⁸ Had there been no foreign patenting in the UK and no foreign markets for UK technology, the dependent and independent variables would have been on a consistent basis. As it is, x denotes the stock of newly patented inventions in the country and not the amount of domestic invention. The variable RD/x is therefore an inappropriate measure of the average cost per invention as RD relates to UK expenditure and x to the total of UK and foreign inventions patented in the UK. Thus Equation 3 is a hybrid, explaining in part, the output of inventions in the UK, but also the transfer of technology from abroad to the UK. It works reasonably well statistically because those variables that are important in explaining the inducement to UK inventors also influence the profitability of the transfer of technology to the UK,⁹ and, to a lesser extent, the profitability of inventive activity carried out abroad. While the global incentives for inventive activity form an interesting avenue for research, this was not the original intention of the Stoneman model.

⁸ There have been significant variations in the ratio of domestic to foreign patents and the 1949 Act may have also affected the relative numbers of domestic and foreign produced inventions patented in the UK.

⁹ See, for example, Schiffl and Kitti (1978) and Bosworth (1980a, b).

A more useful test of the model is to net out foreign patenting, leaving x_D as a measure of domestic inventive output. The flow of foreign technology into the UK may be explained by a separate (if related) equation, but it clearly also has a role to play as an influence on the incentive to invent amongst UK inventors. Following the arguments adopted concerning the competition amongst inventors for scarce outlets for their inventions,

$$R_D = \alpha_1 \left(\frac{I}{Y} \right)^{\xi_1} Y^{\gamma_1} x_D^{\eta_{D1}} x_F^{\eta_{F1}} \quad (13)$$

where subscript 1 denotes the flows associated with domestic inventive activity. Different coefficients appear on x_D and x_F because British and foreign inventions may differ in quality and therefore in their competitiveness for scarce outlets. The quality variables are omitted from Equation 13 at this stage because it is impossible to differentiate P and A for domestic and foreign patents in the published data.

It is not possible to form an estimating equation from Equation 13 alone, by setting $[dR_D/dx_D] = [dC/dx]$, because x_F is not exogeneously determined. While the total supply of new inventions abroad S is unlikely to be greatly affected by the demand pull position of the British economy, or by competition from British inventors, the number of those inventions finding their way into the British economy is affected by these variables,

$$R_F = \alpha_2 \left(\frac{I}{Y} \right)^{\xi_2} Y^{\gamma_2} x_D^{\eta_{D2}} x_F^{\eta_{F2}} S^{\psi_2} \quad (14)$$

where subscript 2 denotes the flows associated with foreign inventive activity. S is included for two reasons: the higher is S the more competition for scarce outlets for inventions in foreign markets and the greater the inducement for foreign inventors to consider the UK as a market; the larger the pool of inventions from which x_F is drawn, the higher the quality of those inventions entering the UK from abroad. Both reasons suggest $\psi > 0$.

The cost equations also require some revision. While the domestic cost equation relates to the amount of inventive activity, $C_D = C(x_D)$, the foreign cost equation relates to the costs of adapting the invention to British market conditions, $C_F = C(x_F)$. Thus, maximizing profits for both groups,

$$\log x_D = \kappa_1 + \frac{1}{\eta_{D1}} \log \left(\frac{dC_D}{dx_D} \right) - \frac{\xi_1}{\eta_{D1}} \log \left(\frac{I}{Y} \right) - \frac{\gamma_1}{\eta_{D1}} \log Y - \frac{\eta_{F1}}{\eta_{D1}} \log x_F \quad (15)$$

and

$$\begin{aligned} \log x_F = & \kappa_2 + \frac{1}{\eta_{F2}} \log \left(\frac{dC_F}{dx_F} \right) - \frac{\xi_2}{\eta_{F2}} \log \left(\frac{I}{Y} \right) \\ & - \frac{\gamma_2}{\eta_{F2}} \log Y - \frac{\eta_{D2}}{\eta_{F2}} \log x_D + \psi_2 \log S \end{aligned} \quad (16)$$

where κ_1 and κ_2 are constant terms. Thus, Equations 15 and 16 form a simultaneous system.

The reduced form for x_D appears as,

$$\begin{aligned} \log x_D = & \kappa + \frac{\eta_{F2}}{\delta} \log \left(\frac{dC_D}{dx_D} \right) + \left(\frac{\xi_2 \eta_{F1} - \xi_1 \eta_{F2}}{\delta} \right) \log \left(\frac{I}{Y} \right) \\ & + \frac{\gamma_2 \eta_{F1} - \gamma_1 \eta_{F2}}{\delta} \log Y - \frac{\eta_{F1}}{\delta} \log \left(\frac{dC_F}{dx_F} \right) - \frac{\eta_{F1} \eta_{F2} \psi_2}{\delta} \log S \end{aligned} \quad (17)$$

where $\delta = (\eta_{D1} \eta_{F2} - \eta_{F1} \eta_{D2})$.

III. DATA, PROBLEMS AND SOURCES

Characteristics of the Boehm data

There are several important points to be made about the sample utilized in the Stoneman (1979) study. First, inventions are allocated to those headings (broadly product areas) where they will be used, rather than to the product areas of the inventing firm. At the fairly high level of industrial aggregation used in the empirical testing, however, this is unlikely to prove a significant problem. Second, the total numbers of acceptances reported in Boehm (1967) differ from those published.¹⁰ Stoneman (1979, p. 391) appears to have corrected the instance where the data sum to only one-third of the published total, but all of the other totals vary by differing amounts. It is also worth pointing out that the patents are divided between primary and secondary headings:¹¹ primary headings deal with the main product area of the invention; subsidiary headings relate to other relevant product areas of less significance. Unless the primary and secondary features can be treated as separate inventions, the use of the total to some extent double counts, and thereby, overstates the number of inventions. The final problem is that the period 1931–1960 includes a change in patent law, with the Patent Act 1949.¹² There is clear historical evidence that other changes in patent law have affected the level of patenting activity.¹³ Fortunately, this change in patent law occurred just after the war and the model was tested including a dummy variable to isolate a shift in the relationship from its pre-war to its post-war level.¹⁴ There is a change in patent law in the sample period used in Section IV and, hence, a dummy variable is included in a number of the regressions in order to test for shifts in the relationship.

Data and sources used in this paper

Stoneman (1979) experienced considerable problems in obtaining R & D data on a compatible basis to the patent acceptance data. This study adopts a time series of more aggregate data, at

¹⁰For example, particularly 1958, 1959 and 1960.

¹¹The use of secondary headings was abandoned after 1963.

¹²For a description, see Boehm (1967).

¹³See Boehm (1967, pp. 14–37) and Schiffel and Kitti (1978, p. 336).

¹⁴It is worth mentioning that the timing of this dummy variable is out by two years in the role of a test of the impact of legislative changes.

broadly manufacturing industry level for the period 1950–75 inclusive. At this somewhat higher level of aggregation, it is possible to construct a consistent time series of constant price R & D expenditure. In addition, it is possible to distinguish between domestic and foreign patenting activity in the UK.

The R & D series was based on data for 1948–70 reported by Schott (1976, p. 93). This series was updated to 1975 by linking it to the expenditure series reported in *Trade and Industry* (1979). The complete series was transformed to constant prices by constructing a price index of the form described by Schott (1976, p. 94).¹⁵ The cost of transferring foreign produced inventions to the UK is a composite of a number of different types of expenditure. Clearly, no evidence from official sources exist about the costs of adapting foreign inventions to UK conditions and the explanatory variable used in the regressions was an index of the costs of patenting activity constructed from the *Annual Report of the Comptroller General of Patents, Designs and Trademarks*.

Data on numbers of patents were taken from the *Annual Report*. The data relate to grants and, thereby, avoid any problems caused by those 'inventions' that reached the acceptance stage but were still not granted patents. The total of patents granted x can be divided into domestic patents x_D and foreign patents x_F . A patent life series P was constructed from renewal data and this is considered in more detail elsewhere (Bosworth, 1973). The 'export' of UK inventions abroad A is proxied by the number of applicants under Section 77, Rule 133¹⁶ of the Patents Act. Such applications represented a means of facilitating patenting outside of the UK. Both sets of data are published in the *Annual Report*. Information about patenting on a worldwide basis S has only been made available since 1963,¹⁷ and, even then, the information was incomplete. In this study, therefore, the number of patents taken out in the USA is taken as a measure of S .

Output and investment per unit of output reflect the demand pull influences. Output and investment data, in constant prices, are taken from *National Income and Expenditure*. It proved possible to test measures based on all industries and services, as against the manufacturing sector. While the former is somewhat wider than the industrial coverage of the patent system, the latter is somewhat too narrow. In practice, it makes little or no difference to the results which of the alternatives is adopted, although, on balance, the results based on manufacturing are marginally superior and these are reported below.

The discount rate is assumed to be a function of minimum lending rate¹⁸ r and the availability of internally generated funds for discretionary investments Π_E . The latter is not, therefore, simply accounting profit Π_A but is constructed as $\Pi_E = \Pi_A + RD + ADV$.¹⁹ Minimum lending rate is published in the *Bank of England Quarterly Bulletin*. Accounting profits are taken as company gross trading profits, published in *National Income and Expenditure*. RD data are

¹⁵ The price index was constructed using: labour cost data (for technical, clerical and administrative workers) from the *Historical Abstract of Labour Statistics* (1971) and *Labour Yearbooks*; indices for plant and machinery, materials, land and buildings, taken from the *Monthly Digest of Statistics*.

¹⁶ Previously Rule 135.

¹⁷ World Intellectual Property Organisation, *Industrial Property*, BIRPI, WIPO, Geneva (published annually).

¹⁸ MLR succeeded bank rate in October 1972.

¹⁹ See, for example, the work of Grabowski and Mueller (1978).

discussed above and advertising data are taken from Critchley (1973), Dunbar (1976) and *Financial Times* (1980). Π_E is constructed in current prices and then deflated by the GDP deflator published in *National Income and Expenditure*.

IV. EMPIRICAL RESULTS

The initial testing of the various hypotheses on the time series data described in Section III presented a number of econometric problems, the most serious of which was the presence of autocorrelation. However, the more 'sophisticated' models based on the reduced form equations did not suffer from this particular problem, although the introduction of additional explanatory variables led to multicollinearity. All of the results reported below exclude the Patent Law dummy discussed in Section III relating to the one major change that took place within the sample period adopted in this study, because in no instance was it statistically significant or did its inclusion add significantly to the overall explanatory power of the model. Given the discussion of the final part of Section II, all of the regression results reported in this section have x_D as the dependent variable.

The tests on the original model, as specified in Equation 8, gave result (1).

Result 1. *Cochrane–Orcutt two-step method* $\rho = 0.55$

Specification: Equation 3

$$\log x_t = 13.297 + 0.638 \log \left(\frac{I}{Y} \right)_{t-1} - 0.276 \log Y_{t-1} - 0.0213 \log \left(\frac{RD_{t-3}}{x_t} \right)$$

(8.970) (3.142) (1.744) (0.481)

$$\bar{R}^2 = 0.7256, F_{(3,19)} = 20.3895, \text{s.e. reg.} = 0.0385, \text{DW} = 1.6296.$$

(Figures in parentheses are absolute t -values. s. e. reg. is standard error of the regression. DW is Durbin–Watson statistic.)

The initial specification reported in Result 1 had severe auto-correlation which was not wholly resolved by the use of the Cochrane–Orcutt (1949) and Hildreth–Lu (1960) techniques. Thus, the result reported is based on the crude Cochrane–Orcutt two-step method to overcome auto-correlation (see Stewart and Wallis, 1981). While the I/Y and RD/x variables have coefficients with the anticipated signs, Y does not. In addition the cost per invention variable has a coefficient that is insignificantly different from zero. The opposite signs of the two 'demand-pull' variables is a consistent feature throughout the alternative specifications tested. The lag structure of this equation was examined in some depth. A consistent picture of a three period lag between R & D activity and patent grants emerged. On the other hand, a one period lag emerged between the 'demand-pull' variables and patenting activity. These lags carry through in all of the regression work undertaken on the alternative specifications of the model.

The addition of Z , to allow for the expected life of the patented invention, corresponding with the revenue function of Equation 5, yields Result 2.

Result 2. *Cochrane–Orcutt two-step method* $\rho = 0.46$

Specification: Equation 5

$$\begin{aligned} \log x_t = & 12.273 + 0.652 \log \left(\frac{I}{Y} \right)_{t-1} - 0.191 \log Y_{t-1} - 0.0151 \log \left(\frac{RD_{t-3}}{x_t} \right) \\ & (5.225) \quad (2.960) \quad (0.870) \quad (-0.352) \\ & + 0.105 \log Z_t \\ & (0.654) \end{aligned}$$

$$\bar{R}^2 = 0.7594, F_{(4,18)} = 14.2066, \text{ s.e. reg.} = 0.0398, \text{ DW} = 1.678.$$

Again this result is based on the application of the Cochrane–Orcutt two-step method in an attempt to remove autocorrelation. The results are only slightly different from those of Result 1 above, with the existing variables maintaining their previous signs and Z appearing with an anticipated positive sign. The coefficient on Z is, however, insignificantly different from zero at the 10 % level. Clearly, this result is not wholly acceptable as several of the variables, including Z , have insignificant coefficients. A possible reason is the presence of multicollinearity, as the F statistic shows an acceptably good overall fit.

Result 3 corresponds with the revenue function of Equation 7. In this equation P and r appear as separate explanatory variables and Π_E is also included with the same period as x . The coefficient on r has the anticipated negative sign but the coefficient on P has an unexpected negative sign. Neither coefficient is significant at the 10 % level or higher, although the coefficient on r comes very close to being significant at this level. Interestingly, the coefficients on Y and RD/x both change signs: while the coefficient on Y now has the anticipated sign, the coefficient on RD/x has a perverse sign. Neither coefficient is significantly different from zero. It is worth adding that the lack of robustness of a number of coefficients in the regression probably reflects the impact of multicollinearity.

Result 3. *Cochrane–Orcutt two-step method* $\rho = 0.31$

Specification: Equation 7

$$\begin{aligned} \log x_t = & 11.718 + 0.438 \log \left(\frac{I}{Y} \right)_{t-1} + 0.0567 \log Y_{t-1} + 0.0174 \log \left(\frac{RD_{t-3}}{x_t} \right) \\ & (6.846) \quad (2.322) \quad (0.321) \quad (0.575) \\ & - 0.105 \log P_t - 0.0742 \log r_t - 0.187 \log \Pi_E \\ & (0.789) \quad (1.648) \quad (2.679) \end{aligned}$$

$$\bar{R}^2 = 0.8129, F_{(6,16)} = 16.9346, \text{ s.e. reg.} = 0.0317, \text{ DW} = 1.6928.$$

The coefficient on Π_E in Result 3 takes a negative sign and is significant at the 1 % level. It appears to be consistent with the argument that high profits at the time of patenting tend to depress the propensity to patent or to innovate, as discussed in Section II. While the results are not reported in detail, exploration of the impact of changing the lag on Π_E indicates that its sign changes (although its coefficient is not significantly different from zero) by the time it is of the same period as R & D expenditure. This is consistent with the ‘internally generated funds argument’, where low profitability adversely affects the amount of inventive activity.

The inclusion of P has, to some extent, already allowed for variations in the quality of inventive output. In addition, however, the theory suggested that the degree of commercial applicability can be further allowed for by the inclusion of A , the extent to which UK inventors take out patents in other countries. Therefore, Equation 10 has been tested, giving Results 4 and 5 below.

Result 4. *Cochrane–Orcutt two-step method* $\rho = 0.45$

Specification: Equation 10

$$\begin{aligned} \log x_t = & 13.754 + 0.699 \log \left(\frac{I}{Y} \right)_{t-1} - 0.460 \log Y_{t-1} - 0.0538 \log \left(\frac{RD_{t-3}}{x_t} \right) \\ & (5.99) \quad (3.409) \quad (1.89) \quad (1.22) \\ & + 0.190 \log Z_t + 0.239 \log A_t \\ & (0.803) \quad (2.01) \\ \bar{R}^2 = & 0.7475, F_{(5,17)} = 14.02, \text{s.e. reg.} = 0.0369, \text{DW} = 1.960. \end{aligned}$$

Result 5. *Ordinary least squares*

Specification based on Equation 10

$$\begin{aligned} \log x_t = & 17.017 + 1.110 \log \left(\frac{I}{Y} \right)_{t-1} - 0.910 \log Y_{t-1} - 0.085 \log \left(\frac{RD_{t-3}}{x_t} \right) \\ & (5.90) \quad (4.20) \quad (2.88) \quad (1.67) \\ & + 0.286 \log P_t + 0.038 \log i_t + 0.452 \log A_t \\ & (1.23) \quad (0.56) \quad (2.45) \\ \bar{R}^2 = & 0.7343, F_{(6,17)} = 7.8322, \text{s.e. reg.} = 0.0533, \text{DW} = 1.669. \end{aligned}$$

Result 4 includes the Z variable, while Result 5 separates this variable into its P and i components. The removal of autocorrelation in Result 4 still requires the application of the Cochrane–Orcutt two-step method. The inclusion of A into Result 5, however, removes autocorrelation and so it was possible to apply ordinary least squares (OLS). This may indicate that autocorrelation in the earlier tests was due, in part, to the misspecification of the equation. These results are more acceptable than their earlier counterparts, although some parameter estimates are still insignificantly different from zero. Some worry still surrounds these results because single equation techniques were used to test what are in effect ‘quasi-reduced form’ equations. While these are not structural equations, neither are they reduced forms of the type represented by Equation 12, which we report on below.

The tests on the reduced form of this system of equations produced the most interesting results. Autocorrelation was not present and the specification was tested using OLS. The coefficient on the variable i proved to be insignificant in all of the tests of the reduced form and is omitted from the results reported here. The most basic version of the reduced form equation is given in Result 6.

Result 6. *Ordinary least squares*

Specification: Equation 12

$$\log x_t = 11.130 + 0.852 \log \left(\frac{I}{Y} \right)_{t-1} - 0.195 \log Y_{t-1} \\ (10.418) \quad (5.554) \quad (1.853) \\ - 0.217 \log \left(\frac{RD_{t-3}}{x_t} \right) + 0.084 \log \left(\frac{RD_t}{A_t} \right) + 0.175 \log \left(\frac{RD_{t-2}}{P_t} \right) \\ (4.233) \quad (2.475) \quad (2.974)$$

$$\bar{R}^2 = 0.8644, F_{(5,18)} = 30.3314, \text{ s.e. reg.} = 0.0327, \text{ DW} = 1.8994.$$

All the coefficients are significantly different from zero at the 10% level or higher. While the coefficient on Y retains its perverse sign, all the remaining signs are consistent with *a priori* expectations. It is particularly interesting that the various dimensions of R & D costs appear to play a significant role in explaining x_t .²⁰ It is also interesting that, when lags are allowed to vary with the dimension involved, different lags are indicated. Adding current (i.e. period t) economic profits to the specification rendered the variable Y insignificant, without this additional variable adding significantly to the overall explanatory power of the model.

While the overall result at first sight appears to be acceptable by normal statistical criteria, a closer examination does reveal at least one problem. Comparison of the coefficients on the three R & D cost terms enables a value for η to be isolated. It suggests, however, that an increase in the number of inventions actually leads to an increase in the average revenue from invention. This result is clearly difficult to interpret within the confines of a model that asserts that inventors compete for scarce outlets for their inventions. It might be explained where the marginal costs of invention are increasing disproportionately quickly with the level of inventive activity, a situation that is difficult to reconcile with the constant marginal cost assumption imposed on the model by data limitations. While it is possible to search for an explanation along these lines, the unexpected sign of η is more likely to be the result of a continued misspecification of the model or parameter instability due to multicollinearity. The value of η estimated in this way turns out to be sensitive to quite small changes in the relative values of the three parameters in this specification.

Respecification of the model to reflect the foreign and domestic dimensions discussed in the final part of Section II without distinguishing the quality and quantity dimensions discussed above is equally likely to result in a misspecified model. It was decided, therefore, to move directly to a specification that incorporates all of these dimensions. The extension of the model to examine not only the interaction of domestic and foreign inventions (shown as Equation 17) but also the quality of inventions (shown as Equation 12), is reported as Result 7. The lags on the various cost of invention variables again mirror those isolated in Result 6, although in this instance, RD/P has the same lag structure as RD/x . From a statistical point of view, the results appear to be acceptable: there are no problems of autocorrelation, a high overall explanatory power and significant coefficients on all of the explanatory variables.

²⁰ A certain degree of licence has been taken in including all three cost of invention variables. In particular, the RD/P variable uses an estimate of patent life that is constructed from data for all inventions (i.e. both domestic and foreign).

Result 7. Ordinary least squares

Adapted version of Equation 17

$$\begin{aligned}
\log x_t = & 7.580 + 0.260 \log \left(\frac{I}{Y} \right)_{t-1} - 0.222 \log Y_{t-1} \\
& (7.591) \quad (1.748) \quad (1.956) \\
& - 0.502 \log \left(\frac{RD_{t-3}}{x_t} \right) + 0.0952 \log \left(\frac{RD_t}{A_t} \right) \\
& (7.325) \quad (2.735) \\
& + 0.385 \log \left(\frac{RD_{t-3}}{P_t} \right) + 0.199 \log \left(\frac{dC_F}{dx_F} \right)_t + 0.217 \log S_t \\
& (5.669) \quad (1.765) \quad (1.765)
\end{aligned}$$

$$\bar{R}^2 = 0.9352, F_{(7,16)} = 48.4312, \text{ s.e. reg.} = 0.0226, \text{ DW} = 2.2644.$$

There is no longer any reason to expect a positive coefficient on the Y variable. The sign of the observed coefficient turns on the relative sensitivity of domestic and foreign inventors to demand conditions and competition for inventive outlets within the UK. A negative sign on Y might, for example, arise where: foreign revenues from using inventions in the UK are more sensitive than domestic to competition from other foreign inventors (i.e. $|\eta_{F2}| > |\eta_{F1}|$) other things being equal; foreign revenues from using inventions in the UK are less sensitive than domestic to variations in UK output but more sensitive to variations in UK competition (i.e. $\gamma_2 < \gamma_1$ and $|\eta_{D2}| > |\eta_{D1}|$) other things equal. The coefficient on Y can be written as $(\gamma_2\eta_{F1} - \gamma_1\eta_{F2})/(\eta_{D1}\eta_{F2} - \eta_{F1}\eta_{D2})$. Bearing in mind that in each case we anticipate that $\eta < 0$, then we can examine a number of specific values of where the overall coefficient will take a negative sign. (1) If, for example, $\gamma_1 = \gamma_2 = \gamma$ and $\eta_{D1} = \eta_{D2}$, then the overall coefficient is negative if $|\eta_{F2}| > |\eta_{F1}|$. (2) If, on the other hand, $\eta_{F1} = \eta_{F2} = \eta_F$, then the coefficient is negative if $\gamma_2 < \gamma_1$ and $|\eta_{D2}| > |\eta_{D1}|$. (3) Again, on the assumption that $\eta_{F1} = \eta_{F2} = \eta_F$, the coefficient is negative if $\gamma_1 < \gamma_2$ and $|\eta_{D2}| < |\eta_{D1}|$. Case (1) alone seems unlikely to be the explanation, because, under analogous assumptions ($\xi_1 = \xi_2 = \xi$ and $\eta_{D1} = \eta_{D2} = \eta_D$) the coefficient on the I/Y term would also be negative. The coexistence of a negative coefficient on Y and positive coefficient on I/Y appears to require that foreign revenues from invention are less sensitive than domestic revenues to UK output but more sensitive to UK I/Y or vice versa. If, for example, $\eta_{F1} = \eta_{F2} = \eta_F$ and, say $\gamma_2 < \gamma$ and $|\eta_{D2}| > |\eta_{D1}|$, then $\xi_2 > \xi_1$.

V. CONCLUSIONS

There appeared to be strong theoretical grounds for respecification of the model proposed by Stoneman (1979). First, the cost per invention variable required careful examination in terms of the lag structure between inputs and outputs in the inventive process. Second, there is always the inherent problem of quality variations when using a simple index of inventive output based on a patent count. Avoidance of this problem, however, resulted in consideration of what dimensions of quality could be captured by the model. Of the various alternatives, the commercial quality of the patented invention (represented by length of patent life and the breadth of international dispersion and use of the patented idea) appeared to be most important. This led to a further

respecification of the model, because work by the inventor on each of these dimensions have cost implications that must be taken into account in deciding the balance between them (e.g. between the absolute number of inventions and their quality). Finally, the earlier use of all patents (independent of country of origin) created an inconsistency between dependent and independent variables in the estimated equation. Acknowledgement of this problem resulted in a further modification to the model to allow the amount of domestic and foreign patenting in the UK to be determined jointly.

Our general conclusion from the empirical work was that respecification of the model in the manner described above resulted in significant improvements both from statistical and interpretive viewpoints. Nevertheless, there were one or two lingering doubts, particularly concerning the robustness of certain of the coefficients because of problems of multicollinearity. Problems of autocorrelation, however, seemed to be resolved by the adoption of the more 'sophisticated' models.

A number of the more detailed aspects of the results appear to be worthy of mention. In particular, it is extremely difficult to interpret the perverse sign of the coefficient on the demand pressure variable Y in the initial specifications. One obvious possibility is that this is the result of multicollinearity between I/Y and Y , although it is intriguing why both variables tend to retain their statistical significance. This problem may have been resolved in the reduced form equations where the coefficient on this variable is a composite of a number of parameters from the structural equations.

The lag structure of the cost of invention variables is an interesting feature of the model. On the RD/x variable, a three-period lag is consistently isolated. This corresponds closely with our *a priori* expectations in the sense that our discussion of the patent data indicated at least a two-year lag between application and grant. The model appears to indicate that the period between R & D expenditure and application for a patent is generally of the order of one year.²¹ Clearly, if this is the case then the majority of R & D programmes give rise to fairly immediate results. This would be consistent with the view that the bulk of patented inventions are associated with fairly minor inventions and modifications. It would also be the case, however, where research programmes build up to a climax during the final year or so before patenting.

It would seem probable that the likely longevity of the invention will be determined at broadly the same time that the underlying principle of the invention is established, although clearly additional later research may be devoted to modifying the invention even after an application for a patent has been made. The adaptation of the invention to other countries, associated with RD/A , is likely to take place at a later stage than the research into the underlying principle and probably after the expected lifespan in the UK has been established. The lag structures isolated in the results reported in Section IV are consistent with these *a priori* expectations, indicating a three-year lag between RD and x , a two- or three-year lag between RD and P , but no appreciable lag between RD and A .

There is a certain amount of evidence that high profits at the time of R & D expenditure, stimulate the amount of R & D and thereby the size of inventive output, according to the internally generated funds hypothesis. There is, however, also evidence that higher profits at the

²¹ This has been further confirmed by work based on a polynomial lag structure, which is not reported in detail here.

time of the innovation decision appear to suppress the propensity to patent. It might be the case that firms may be more reluctant to introduce new ideas when existing products and processes are sufficiently profitable. It may also indicate that their existing barriers to entry (in particular their existing control over technical knowhow in the form of patents) may be sufficiently strong and the benefits of further patents are relatively low.

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